

Module 2: Geometry, Kinematics, Statics, Dynamics

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6th May, 2026

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1 Physical Vectors and Frames

Definition 1.1. A *point* has zero size and zero mass. A *particle* has zero size and nonzero mass.

Points and particles have position *relative* to other points and particles.

Definition 1.2. A *body* is a collection of particles.

Definition 1.3. A *physical vector* has a magnitude and direction.

We will develop kinematics and dynamics using various physical vectors such as physical position vector, physical velocity vector, physical acceleration vector, physical angular velocity vector, physical force vector, physical moment vector, etc.

Example 1.1. Let x and y be two points. Then, $\vec{r}_{y/x}$ is the physical position vector from x to y .

Fact 1.1. Although x and y have locations relative to each other, the physical vector $\vec{r}_{y/x}$ has no location because $\vec{r}_{y/x}$ always represents the location of y relative to x , irrespective of where x is.

1.1 Angles

Let $\vec{x}$ and $\vec{y}$ be two physical vectors. Let θ be the angle between $\vec{x}$ and $\vec{y}$.

Definition 1.4. An angle $\theta \in \mathbb{R}$ is an *unwrapped angle*.

Definition 1.5. An angle $\theta \in (-\pi, \pi]$ is a *wrapped angle*.

Definition 1.6. An angle $\theta \in [0, \pi]$ is a *short-way angle*.

Let $\theta_{\vec{x}/\vec{y}} = \theta_{\vec{y}/\vec{x}} \in [0, \pi]$ denote the short-way angle between the physical vectors $\vec{x}$ and $\vec{y}$.

Definition 1.7. The inner product $\vec{x} \cdot \vec{y}$ is defined by

$$\vec{x} \cdot \vec{y} \triangleq |\vec{x}| |\vec{y}| \cos \theta_{\vec{x}/\vec{y}}. \quad (1)$$

We define

$$\vec{x}' \vec{y} \triangleq \vec{x} \cdot \vec{y}, \quad (2)$$

where the *physical covector* $\vec{x}'$ operates on the physical vector $\vec{y}$ to produce the real scalar $\vec{x} \cdot \vec{y}$. Note that $\vec{x}' \vec{y} = \vec{y}' \vec{x}$. The physical covector $\vec{x}'$ is the *coform* of the physical vector $\vec{x}$. We define $(\vec{x}')' \triangleq \vec{x}$.

Definition 1.8. Let $\vec{x}$ and $\vec{y}$ be nonzero physical vectors that are not parallel so that $\theta_{\vec{y}/\vec{x}} \in (0, \pi)$. The *unit angle vector* $\hat{\theta}_{\vec{y}/\vec{x}}$ of $\vec{y}$ relative to $\vec{x}$ is the unit dimensionless physical vector orthogonal to both $\vec{x}$ and $\vec{y}$ whose direction is determined by the right-hand rule with the fingers curled from $\vec{x}$ to $\vec{y}$ through the short-way angle $\theta_{\vec{y}/\vec{x}}$ between $\vec{x}$ and $\vec{y}$.

Definition 1.9. Let $\vec{x}$ and $\vec{y}$ be nonzero physical vectors. If $\vec{x}$ and $\vec{y}$ are not parallel, then the *cross product* of $\vec{x}$ and $\vec{y}$ is defined as

$$\vec{x} \times \vec{y} \triangleq |\vec{x}| |\vec{y}| (\sin \theta_{\vec{y}/\vec{x}}) \hat{\theta}_{\vec{y}/\vec{x}}. \quad (3)$$

1.2 Frame

Definition 1.10. A *frame* is a set of three mutually orthogonal physical vectors.

Frames are used to define directions in three-dimensional space. For simplicity, we will use unit and dimensionless mutually orthogonal physical vectors, denoted by $\hat{i}_A, \hat{j}_A$, and $\hat{k}_A$, to construct frame F_A . We represent the frame F_A as the row vectrix $F_A = [\hat{i}_A \ \hat{j}_A \ \hat{k}_A]$.

Fact 1.2. Since physical vectors have no location, a frame has no location.

Consider a right handed frame F_A . The frame axes are denoted by $\hat{i}_A, \hat{j}_A, \hat{k}_A$.

Fact 1.3. The frame axes satisfy the following dot products.

$$\hat{i}'_A \hat{i}_A = \hat{j}'_A \hat{j}_A = \hat{k}'_A \hat{k}_A = 1, \quad (4)$$

$$\hat{i}'_A \hat{j}_A = \hat{i}'_A \hat{k}_A = \hat{j}'_A \hat{i}_A = \hat{j}'_A \hat{k}_A = \hat{k}'_A \hat{i}_A = \hat{k}'_A \hat{j}_A = 0. \quad (5)$$

Fact 1.4. The frame axes satisfy the following cross products.

$$\hat{i}_A \times \hat{j}_A = \hat{k}_A, \quad \hat{j}_A \times \hat{k}_A = \hat{i}_A, \quad \hat{k}_A \times \hat{i}_A = \hat{j}_A, \quad (6)$$

$$\hat{i}_A \times \hat{k}_A = -\hat{j}_A, \quad \hat{k}_A \times \hat{j}_A = -\hat{i}_A, \quad \hat{j}_A \times \hat{i}_A = -\hat{k}_A, \quad (7)$$

$$\hat{i}_A \times \hat{i}_A = \hat{j}_A \times \hat{j}_A = \hat{k}_A \times \hat{k}_A = 0. \quad (8)$$

Fact 1.5. Let F_A and F_B be two frames. Then, there exists a physical vector $\hat{n}_{B/A}$ and the angle $\theta_{B/A}$ such that F_B is obtained by rotating F_A around $\hat{n}_{B/A}$ by angle $\theta_{B/A}$.

1.3 Resolving physical vectors

Let F_A be a frame, and let $\vec{x} = x_1 \hat{i}_A + x_2 \hat{j}_A + x_3 \hat{k}_A$ be a physical vector. Then, $\vec{x}|_A$ is the physical vector $\vec{x}$ resolved in F_A . In fact, $\vec{x}|_A$ is the *mathematical vector* defined by

$$\vec{x}|_A \triangleq \begin{bmatrix} \hat{i}'_A \vec{x} \\ \hat{j}'_A \vec{x} \\ \hat{k}'_A \vec{x} \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad (9)$$

where x_1, x_2 , and x_3 are the *components* of the physical vector $\vec{x}$ resolved in F_A . The components of $\vec{x}$ resolved in F_A are the *Cartesian coordinates* of $\vec{x}$ in F_A .

Similarly,

$$\vec{x}'|_A \triangleq \vec{x}|_A^T = [x_1 \ x_2 \ x_3], \quad (10)$$

Fact 1.6. The frame axes resolved in F_A are

$$\hat{i}_A|_A = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \hat{j}_A|_A = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \hat{k}_A|_A = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}. \quad (11)$$

1.4 Physical vector operations

Let $\vec{x} = x_1 \hat{i}_A + x_2 \hat{j}_A + x_3 \hat{k}_A$ and $\vec{y} = y_1 \hat{i}_A + y_2 \hat{j}_A + y_3 \hat{k}_A$. Then,

$$\vec{x}|_A = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \vec{y}|_A = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}, \quad (12)$$

$$\vec{x} + \vec{y} = (x_1 + y_1) \hat{i}_A + (x_2 + y_2) \hat{j}_A + (x_3 + y_3) \hat{k}_A, \quad (13)$$

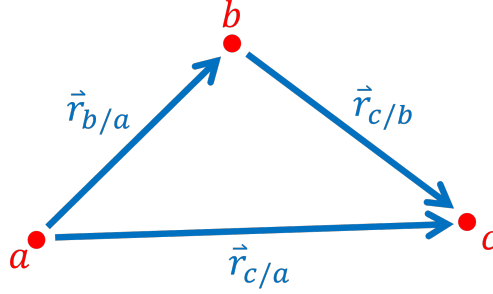
$$\vec{x} \times \vec{y} = (x_2 y_3 - x_3 y_2) \hat{i}_A + (x_3 y_1 - x_1 y_3) \hat{j}_A + (x_1 y_2 - x_2 y_1) \hat{k}_A, \quad (14)$$

$$\vec{x}' \vec{y} = \vec{x} \bullet \vec{y} = x_1 y_1 + x_2 y_2 + x_3 y_3, \quad (15)$$

1.5 Slash-and-split property

Some physical vectors satisfy the slash-and-split property. Let $\vec{r}_{b/a} = x_1\hat{i}_A + x_2\hat{j}_A + x_3\hat{k}_A$ and $\vec{r}_{c/b} = y_1\hat{i}_A + y_2\hat{j}_A + y_3\hat{k}_A$. Then,

$$\vec{r}_{c/a} = \vec{r}_{c/b} + \vec{r}_{b/a} = (x_1 + y_1)\hat{i}_A + (x_2 + y_2)\hat{j}_A + (x_3 + y_3)\hat{k}_A. \quad (16)$$



1.6 Physical Matrices

Let $\vec{x}_1, \dots, \vec{x}_n$ and $\vec{y}_1, \dots, \vec{y}_n$ be physical vectors. Then,

$$\vec{M} \triangleq \sum_{i=1}^n \vec{x}_i \vec{y}_i' \quad (17)$$

is a *physical matrix*. A physical matrix is a second-order tensor.

Let $\vec{M} = \vec{x} \vec{y}'$, and $\vec{N} = \vec{y} \vec{x}'$. Then,

$$\vec{M} \vec{N} = \vec{y} \vec{x}' \vec{x} \vec{y}' = (\vec{x}' \vec{x}) \vec{y} \vec{y}', \quad (18)$$

$$\vec{M} \vec{x} = \vec{x} \vec{y}' \vec{x} = (\vec{y}' \vec{x}) \vec{x}. \quad (19)$$

Let $\vec{x} = x_1\hat{i}_A + x_2\hat{j}_A + x_3\hat{k}_A$ and $\vec{y} = y_1\hat{i}_A + y_2\hat{j}_A + y_3\hat{k}_A$. Then,

$$\vec{x} \vec{y}' = (x_1\hat{i}_A + x_2\hat{j}_A + x_3\hat{k}_A)(y_1\hat{i}_A' + y_2\hat{j}_A' + y_3\hat{k}_A') \quad (20)$$

$$\begin{aligned} &= (x_1y_1)\hat{i}_A\hat{i}_A' + (x_1y_2)\hat{i}_A\hat{j}_A' + (x_1y_3)\hat{i}_A\hat{k}_A' \\ &+ (x_2y_1)\hat{j}_A\hat{i}_A' + (x_2y_2)\hat{j}_A\hat{j}_A' + (x_2y_3)\hat{j}_A\hat{k}_A' \\ &+ (x_3y_1)\hat{k}_A\hat{i}_A' + (x_3y_2)\hat{k}_A\hat{j}_A' + (x_3y_3)\hat{k}_A\hat{k}_A'. \end{aligned} \quad (21)$$

Let $\vec{x}$ be a physical vector. Then, for all physical vectors $\vec{y}$, the *physical cross product matrix* $\vec{M} \triangleq \vec{x}^\times$ is defined by

$$\vec{M} \vec{y} = \vec{x}^\times \vec{y} \triangleq \vec{x} \times \vec{y}. \quad (22)$$

Definition 1.11. Let

$$x|_A = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}. \quad (23)$$

Then,

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}^\times = \vec{x}|_A^\times = \begin{bmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{bmatrix} \quad (24)$$

is called the supercross matrix.

2 Orientation Matrix

Let $\vec{x}$ be a physical vector. Let F_A and F_B be two frames. Then, the mathematical vectors $\vec{x}|_A$ and $\vec{x}|_B$ are related by

$$\vec{x}|_B = \mathcal{O}_{B/A} \vec{x}|_A, \quad (25)$$

where the 3×3 positive-definite matrix $\mathcal{O}_{B/A}$ is called the *orientation matrix*.

Fact 2.1. The orientation matrix satisfies

$$\mathcal{O}_{A/B} = \mathcal{O}_{B/A}^{-1}, \quad (26)$$

$$\mathcal{O}_{C/A} = \mathcal{O}_{C/B} \mathcal{O}_{B/A}. \quad (27)$$

Fact 2.2. The orientation matrix is orthogonal, that is,

$$\mathcal{O}_{B/A}^T \mathcal{O}_{B/A} = I_3, \quad (28)$$

which implies that

$$\mathcal{O}_{B/A}^{-1} = \mathcal{O}_{B/A}^T = \mathcal{O}_{A/B}. \quad (29)$$

Fact 2.3. The frame axes satisfy

$$\begin{bmatrix} \hat{i}_B \\ \hat{j}_B \\ \hat{k}_B \end{bmatrix} = \mathcal{O}_{B/A} \begin{bmatrix} \hat{i}_A \\ \hat{j}_A \\ \hat{k}_A \end{bmatrix}. \quad (30)$$

The entries of $\mathcal{O}_{B/A}$ are the cosines of the angles between pairs of vectors in frames F_B and F_A . Consequently, $\mathcal{O}_{B/A}$ is a *direction cosine matrix*.

Fact 2.4. Let F_A and F_B be frames. Then,

$$\mathcal{O}_{B/A} = \begin{bmatrix} \cos \theta_{\hat{i}_B/\hat{i}_A} & \cos \theta_{\hat{i}_B/\hat{j}_A} & \cos \theta_{\hat{i}_B/\hat{k}_A} \\ \cos \theta_{\hat{j}_B/\hat{i}_A} & \cos \theta_{\hat{j}_B/\hat{j}_A} & \cos \theta_{\hat{j}_B/\hat{k}_A} \\ \cos \theta_{\hat{k}_B/\hat{i}_A} & \cos \theta_{\hat{k}_B/\hat{j}_A} & \cos \theta_{\hat{k}_B/\hat{k}_A} \end{bmatrix}. \quad (31)$$

Fact 2.5. Let F_A and F_B be frames. Then,

$$\mathcal{O}_{B/A} = (\cos \theta_{B/A}) I_3 + (1 - \cos \theta_{B/A}) n_{B/A} n_{B/A}^T - (\sin \theta_{B/A}) n_{B/A}^\times. \quad (32)$$

2.1 Euler orientation matrices

2.1.1 Rotation around $\hat{i}_A$ by angle θ

$$F_A \xrightarrow[1]{\theta} F_B = F_A \xrightarrow[\hat{i}_A]{\theta} F_B \implies \mathcal{O}_{B/A} = \mathcal{O}_1(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{bmatrix}. \quad (33)$$

2.1.2 Rotation around $\hat{j}_A$ by angle θ

$$F_A \xrightarrow[2]{\theta} F_B = F_A \xrightarrow[\hat{j}_A]{\theta} F_B \implies \mathcal{O}_{B/A} = \mathcal{O}_2(\theta) = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix}. \quad (34)$$

2.1.3 Rotation around $\hat{k}_A$ by angle θ

$$F_A \xrightarrow[3]{\theta} F_B = F_A \xrightarrow[\hat{k}_A]{\theta} F_B \implies \mathcal{O}_{B/A} = \mathcal{O}_3(\theta) = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (35)$$

2.2 Sequence of Euler Rotations

For example, consider the sequence of rotations

$$F_A \xrightarrow[3]{\psi} F_B \xrightarrow[2]{\theta} F_C \xrightarrow[1]{\phi} F_D. \quad (36)$$

Then,

$$\mathcal{O}_{B/A} = \mathcal{O}_3(\psi), \quad (37)$$

$$\mathcal{O}_{C/B} = \mathcal{O}_2(\theta), \quad (38)$$

$$\mathcal{O}_{D/C} = \mathcal{O}_1(\phi), \quad (39)$$

which implies that

$$\mathcal{O}_{D/A} = \mathcal{O}_{D/C} \mathcal{O}_{C/B} \mathcal{O}_{B/A} = \mathcal{O}_1(\phi) \mathcal{O}_2(\theta) \mathcal{O}_3(\psi).$$

and

$$\mathcal{O}_{A/D} = \mathcal{O}_{A/B} \mathcal{O}_{B/C} \mathcal{O}_{C/D} = \mathcal{O}_{B/A}^T \mathcal{O}_{C/B}^T \mathcal{O}_{D/C}^T = \mathcal{O}_3^T(\psi) \mathcal{O}_2^T(\theta) \mathcal{O}_1^T(\phi). \quad (40)$$

Furthermore,

$$\hat{i}_D|_A = \mathcal{O}_{A/D} \hat{i}_D|_D = \mathcal{O}_3^T(\psi) \mathcal{O}_2^T(\theta) \mathcal{O}_1^T(\phi) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}. \quad (41)$$

3 Frame Derivatives

Consider frames F_A and F_B and let $\vec{x} = x_1 \hat{i}_B + x_2 \hat{j}_B + x_3 \hat{k}_B$. Then

$$\overset{A\bullet}{\vec{x}} = \dot{x}_1 \hat{i}_B + x_1 \overset{A\bullet}{\hat{i}_B} + \dot{x}_2 \hat{j}_B + x_2 \overset{A\bullet}{\hat{j}_B} + \dot{x}_3 \hat{k}_B + x_3 \overset{A\bullet}{\hat{k}_B}, \quad (42)$$

$$\overset{B\bullet}{\vec{x}} = \dot{x}_1 \hat{i}_B + \dot{x}_2 \hat{j}_B + \dot{x}_3 \hat{k}_B. \quad (43)$$

Note that $\overset{B\bullet}{\hat{i}_B} = \overset{B\bullet}{\hat{j}_B} = \overset{B\bullet}{\hat{k}_B} = 0$.

4 Physical Velocity and Acceleration Vectors

Consider points w, y and frames F_A and F_B . Then, the physical vector representing the velocity of y relative to w with respect to Frame F_A is

$$\vec{v}_{y/w/A} \triangleq \overset{A\bullet}{\vec{r}}_{y/w}, \quad (44)$$

and the physical vector representing the acceleration of y relative to w with respect to Frame F_A is

$$\vec{a}_{y/w/A} \triangleq \overset{A\bullet}{\vec{v}}_{y/w/A} = \overset{A\bullet\bullet}{\vec{r}}_{y/w}. \quad (45)$$

Fact 4.1. The physical velocity vector and the physical acceleration vector satisfy the slash-and-split rule, that is,

$$\vec{v}_{z/w/A} = \vec{v}_{z/y/A} + \vec{v}_{y/w/A}, \quad (46)$$

$$\vec{a}_{z/w/A} = \vec{a}_{z/y/A} + \vec{a}_{y/w/A}. \quad (47)$$

5 Physical Angular Velocity Vector

Consider frames F_A and F_B . Then, the physical vector representing the angular velocity of F_B relative to F_A is $\vec{\omega}_{B/A}$ and satisfies

$$\vec{\omega}_{B/A} = -\vec{\omega}_{A/B}. \quad (48)$$

Fact 5.1. The physical angular velocity vector satisfies the slash-and-split rule, that is,

$$\vec{\omega}_{C/A} = \vec{\omega}_{C/B} + \vec{\omega}_{B/A}. \quad (49)$$

5.1 Rotation around $\hat{i}_A$ by angle θ

$$F_A \xrightarrow[1]{\theta} F_B = F_A \xrightarrow[\hat{i}_A]{\theta} F_B \implies \vec{\omega}_{B/A} = \dot{\theta} \hat{i}_A. \quad (50)$$

5.2 Rotation around $\hat{j}_A$ by angle θ

$$F_A \xrightarrow[2]{\theta} F_B = F_A \xrightarrow[\hat{j}_A]{\theta} F_B \implies \vec{\omega}_{B/A} = \dot{\theta} \hat{j}_A. \quad (51)$$

5.3 Rotation around $\hat{k}_A$ by angle θ

$$F_A \xrightarrow[3]{\theta} F_B = F_A \xrightarrow[\hat{k}_A]{\theta} F_B \implies \vec{\omega}_{B/A} = \dot{\theta} \hat{k}_A. \quad (52)$$

6 Transport Theorem

Consider frames F_A and F_B , and physical vector $\vec{x}$. Then,

$$\overset{A\bullet}{\vec{x}} = \overset{B\bullet}{\vec{x}} + \vec{\omega}_{B/A} \times \vec{x}. \quad (53)$$

As a consequence of the transport theorem,

$$\overset{A\bullet}{\hat{i}_B} = \vec{\omega}_{B/A} \times \hat{i}_B, \quad (54)$$

$$\overset{A\bullet}{\hat{j}_B} = \vec{\omega}_{B/A} \times \hat{j}_B, \quad (55)$$

$$\overset{A\bullet}{\hat{k}_B} = \vec{\omega}_{B/A} \times \hat{k}_B. \quad (56)$$

7 The Angular Velocity Vector and Euler-Angle Derivatives

Consider the 3-2-1 sequence of rotations

$$F_A \xrightarrow[3]{\psi} F_B \xrightarrow[2]{\theta} F_C \xrightarrow[1]{\phi} F_D. \quad (57)$$

where ψ, θ, ϕ denote the Euler angles for yaw, pitch, and roll, respectively. Then,

$$\vec{\omega}_{D/A} = \vec{\omega}_{D/C} + \vec{\omega}_{C/B} + \vec{\omega}_{B/A} = \dot{\phi} \hat{i}_D + \dot{\theta} \hat{j}_C + \dot{\psi} \hat{k}_B, \quad (58)$$

where $\dot{\phi}, \dot{\theta}, \dot{\psi}$ denote the Euler-angle derivatives corresponding to the roll rate, pitch rate, and yaw rate, respectively. Resolving in F_D yields

$$\begin{aligned}
 \vec{\omega}_{D/A}|_D &= \dot{\phi} \hat{i}_D|_D + \dot{\theta} \hat{j}_C|_D + \dot{\psi} \hat{k}_B|_D, \\
 &= \dot{\phi} e_1 + \dot{\theta} \mathcal{O}_{D/C} \hat{j}_C|_C + \dot{\psi} \mathcal{O}_{D/B} \hat{k}_B|_B, \\
 &= \dot{\phi} e_1 + \dot{\theta} \mathcal{O}_1(\phi) e_2 + \dot{\psi} \mathcal{O}_1(\phi) \mathcal{O}_2(\theta) e_3, \\
 &= [e_1 \quad \mathcal{O}_1(\phi) e_2 \quad \mathcal{O}_1(\phi) \mathcal{O}_2(\theta) e_3] \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}, \\
 &= S(\phi, \theta) \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix},
 \end{aligned} \tag{59}$$

where

$$S(\phi, \theta) \triangleq [e_1 \quad \mathcal{O}_1(\phi) e_2 \quad \mathcal{O}_1(\phi) \mathcal{O}_2(\theta) e_3] = \begin{bmatrix} 1 & 0 & -\sin \theta \\ 0 & \cos \phi & (\sin \phi) \cos \theta \\ 0 & -\sin \phi & (\cos \phi) \cos \theta \end{bmatrix}. \tag{60}$$

Note that $\det(S(\phi, \theta)) = \cos \theta$. In the case where $\det(S(\phi, \theta)) \neq 0$, it follows that

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = S(\phi, \theta)^{-1} \vec{\omega}_{D/A}|_D = \begin{bmatrix} 1 & (\sin \phi) \tan \theta & (\cos \phi) \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & (\sin \phi) \sec \theta & (\cos \phi) \sec \theta \end{bmatrix} \vec{\omega}_{D/A}|_D. \tag{61}$$

8 Statics

8.1 Moments of Mass

Let $\mathcal{B}$ be a body composed of particles $y_1, y_2, \dots, y_\ell$, whose masses are $m_1, m_2, \dots, m_\ell$, respectively.

Definition 8.1. Let w be a point. Then, the **center of mass** c of $\mathcal{B}$ relative to w is

$$\vec{r}_{c/w} \triangleq \frac{1}{m_{\mathcal{B}}} \sum_{i=1}^{\ell} m_i \vec{r}_{y_i/w}, \tag{62}$$

where $m_{\mathcal{B}} \triangleq m_{\mathcal{B}} \triangleq \sum_{i=1}^{\ell} m_i$.

Definition 8.2. The **second moment of mass** of $\mathcal{B}$ relative to a point z is the physical inertia matrix

$$\vec{J}_{\mathcal{B}/z} \triangleq \sum_{i=1}^{\ell} m_i \vec{r}_{y_i/z}^{\times'} \vec{r}_{y_i/z}^{\times} = - \sum_{i=1}^{\ell} m_i \vec{r}_{y_i/z}^{\times 2} = \sum_{i=1}^{\ell} m_i \left(|\vec{r}_{y_i/z}|^2 \vec{I} - \vec{r}_{y_i/z} \vec{r}_{y_i/z}' \right) \tag{63}$$

Fact 8.1. Generalized Parallel axis theorem. Let z be a point. Then,

$$\vec{J}_{\mathcal{B}/z} \triangleq \vec{J}_{\mathcal{B}/c} + m_{\mathcal{B}} \left(|\vec{r}_{z/c}|^2 \vec{I} - \vec{r}_{z/c} \vec{r}_{z/c}' \right). \tag{64}$$

Fact 8.2. Let body $\mathcal{B}$ be composed of bodies $\mathcal{B}_1$ and $\mathcal{B}_2$, such that $\mathcal{B} = \mathcal{B}_1 \cup \mathcal{B}_2$, and let z be a point. Then,

$$\vec{J}_{\mathcal{B}/z} = \vec{J}_{\mathcal{B}_1/z} + \vec{J}_{\mathcal{B}_2/z}. \tag{65}$$

Fact 8.3. Consider frames F_A and F_B , body $\mathcal{B}$ and point z . Then, the physical inertia matrix of $\mathcal{B}$ relative to z , $\vec{J}_{\mathcal{B}/z}$, satisfies

$$\vec{J}_{\mathcal{B}/z} \Big|_A = \mathcal{O}_{A/B} \vec{J}_{\mathcal{B}/z} \Big|_B \mathcal{O}_{B/A}. \tag{66}$$

8.2 Forces

Forces between interacting bodies arise from three types of mechanisms. *Contact* between two bodies gives rise to *reaction forces*. *Mechanical coupling* between two bodies connected by springs, dashpots, and inerters gives rise to *mechanical forces*. *Field coupling* between two bodies due to central gravity, electrostatics, electrodynamics, and magnetism gives rise to *field forces*.

Definition 8.3. Let $\mathcal{B}_1$ and $\mathcal{B}_2$ be bodies. If $\mathcal{B}_1$ is in contact with $\mathcal{B}_2$ at the point x in $\mathcal{B}_1$, then the *reaction force applied to $\mathcal{B}_1$ due to contact at x* is denoted by $\vec{f}_{\mathcal{B}_1/\mathcal{B}_2/x}$.

Fact 8.4. Newton's Third Law (N3L). Let $\mathcal{B}_1$ and $\mathcal{B}_2$ be two bodies and let x be the point of contact between $\mathcal{B}_1$ and $\mathcal{B}_2$. Let $\vec{f}_{\mathcal{B}_1/\mathcal{B}_2/x}$ be the reaction force applied to $\mathcal{B}_1$ at x and let $\vec{f}_{\mathcal{B}_2/\mathcal{B}_1/x}$ be the net reaction force applied to $\mathcal{B}_2$ due to contact with $\mathcal{B}_1$. Then,

$$\vec{f}_{\mathcal{B}_1/\mathcal{B}_2/x} = -\vec{f}_{\mathcal{B}_2/\mathcal{B}_1/x}, \quad (67)$$

$$\vec{f}_{\mathcal{B}_1/\mathcal{B}_2} = -\vec{f}_{\mathcal{B}_2/\mathcal{B}_1}. \quad (68)$$

Uniform gravity is an approximation of the force exerted by a massive body such as the Earth on a particle. In particular, the force applied by uniform gravity to the particle y with mass m is $m\vec{g}$, where $\vec{g} = g\hat{k}_A$, where F_A is an inertial frame and $\hat{k}_A$ points toward the massive body. By Newton's second law, the acceleration $\vec{g}$ due to uniform gravity is thus the inertial acceleration of y when no other forces are applied to y , that is,

$$\vec{a}_{y/w/A} = \vec{g}. \quad (69)$$

Fact 8.5. The gravitational force on $\mathcal{B}$ is applied at its center of mass c .

8.3 Moments and Torques

Definition 8.4. Let $\mathcal{B}$ be a body with points $y_1, y_2, \dots, y_\ell$, let forces $\vec{f}_{y_1}, \vec{f}_{y_2}, \dots, \vec{f}_{y_\ell}$ be applied at the corresponding points, and let w be a point. Then, the **moment** applied to $\mathcal{B}$ relative to w due to forces $\vec{f}_{y_1}, \vec{f}_{y_2}, \dots, \vec{f}_{y_\ell}$ is

$$\vec{M}_{\mathcal{B}/w} = \sum_{i=1}^{\ell} \vec{r}_{y_i/w} \times \vec{f}_{y_i}. \quad (70)$$

Fact 8.6. Let z be a point. Then,

$$\vec{M}_{\mathcal{B}/z} = \vec{M}_{\mathcal{B}/w} + \vec{r}_{w/z} \times \vec{f}_{\mathcal{B}}, \quad (71)$$

where

$$\vec{f}_{\mathcal{B}} = \sum_{i=1}^{\ell} \vec{f}_{y_i}. \quad (72)$$

Definition 8.5. The forces $\vec{f}_{y_1}, \vec{f}_{y_2}, \dots, \vec{f}_{y_\ell}$ are called **balanced** if

$$\vec{f}_{\mathcal{B}} = \sum_{i=1}^{\ell} \vec{f}_{y_i} = 0. \quad (73)$$

Fact 8.7. If $\vec{f}_{y_1}, \vec{f}_{y_2}, \dots, \vec{f}_{y_\ell}$ are balanced, then

$$\vec{M}_{\mathcal{B}/w} = \vec{M}_{\mathcal{B}/z}, \quad (74)$$

where w, z are points.

Definition 8.6. The **torque** applied to body $\mathcal{B}$ due to balanced forces $\vec{f}_{y_1}, \vec{f}_{y_2}, \dots, \vec{f}_{y_\ell}$ is

$$\vec{T}_{\mathcal{B}} = \vec{M}_{\mathcal{B}/w}, \quad (75)$$

where w is a point.

8.4 Laws of Statics

Definition 8.7. A body $\mathcal{B}$ is in **static equilibrium** if and only if

1. $\vec{f}_{\mathcal{B}} = 0$.
2. $\vec{T}_{\mathcal{B}} = 0$, that is, $\vec{M}_{\mathcal{B}/z} = 0$ for any point z .

9 Dynamics

Definition 9.1. An *unforced particle* is a particle that has no forces applied to it.

Definition 9.2. The frame F_A is an *inertial frame* if, for all unforced particles y and w ,

$$\overset{A\bullet\bullet}{\vec{r}}_{y/w} = 0. \quad (76)$$

Fact 9.1. Newton's First Law (N1L). There exists an inertial frame.

Fact 9.2. Newton's Second Law (N2L). Let F_A be an inertial frame. Let $\mathcal{B}$ be a rigid body with mass $m_{\mathcal{B}}$, and center of mass c . Let $\vec{f}_{\mathcal{B}}$ be the net force applied to $\mathcal{B}$. Let w be an unforced particle. Then,

$$m_{\mathcal{B}} \vec{a}_{c/w/A} = \vec{f}_{\mathcal{B}}. \quad (77)$$

Fact 9.3. Euler's Equations relative to unforced particle. Let F_A be an inertial frame. Let $\mathcal{B}$ be a rigid body with mass $m_{\mathcal{B}}$. Let F_B be a body-fixed frame. Let w be an unforced particle. Let $\vec{M}_{\mathcal{B}/w}$ be the net moment applied to $\mathcal{B}$ relative to w . Then,

$$\vec{J}_{\mathcal{B}/w} \overset{B\bullet}{\vec{\omega}}_{B/A} + \vec{\omega}_{B/A} \times \vec{J}_{\mathcal{B}/w} \vec{\omega}_{B/A} = \vec{M}_{\mathcal{B}/w}. \quad (78)$$

Fact 9.4. Euler's Equations relative to center of mass. Let F_A be an inertial frame. Let $\mathcal{B}$ be a rigid body with mass $m_{\mathcal{B}}$, and center of mass c . Let F_B be a body-fixed frame. Let $\vec{M}_{\mathcal{B}/c}$ be the net moment applied to $\mathcal{B}$ relative to c . Then,

$$\vec{J}_{\mathcal{B}/c} \overset{B\bullet}{\vec{\omega}}_{B/A} + \vec{\omega}_{B/A} \times \vec{J}_{\mathcal{B}/c} \vec{\omega}_{B/A} = \vec{M}_{\mathcal{B}/c}. \quad (79)$$